## James Ferdinand Combista: Mid-Term Mock Test Question

## Question 1: (Combined, Long Answer, and Medium)

Create a coordinate system that asks for an input `n` from the user for an `n x n` grid and return the coordinates in 2D list format, starting from `(0, 0)` at the bottom left of the grid. The coordinates should increment the `x` values from left to right and move upward along the y-axis. The coordinate system will start at `0` on both axes. If the input is an odd number, replace the tuple in the middle of it with `" ** "`. (There is 1 space on either side.)

**Input:** 3

**Ouput:**

```
[(0, 2), (1, 2), (2, 2)]
[(0, 1),   **  , (2, 1)]
[(0, 0), (1, 0), (2, 0)]
```

**SOLUTION:**
```
def coordinates(n):
    grid = []
    for y in range(n):
        y_axis = []
        for x in range(n):
            t = (x, y)
            if n % 2 == 1 and x == y == n // 2:
                y_axis.append(" ** ")
            else:
                y_axis.append(t)
        grid.append(y_axis)
    return grid[::-1]

n = int(input())
result = coordinates(n)
for row in result:
    print(row)
```

## Question 2: (Nested Loops, Trace the code, Difficult)

What does this print?

```
def someFunction(n):
    lst = []
    for i in range(n):
        row = [j + 1 + i * n for j in range(n)]
        lst.append(row)

    for i in range(len(lst)):
        for j in range(len(lst[i])):
            for k in range(2, int(lst[i][j]**0.5) + 1):
                    if lst[i][j] % k == 0:
                            lst[i][j] = 0
    return lst

modified = someFunction(5)
for row in modified:
    print(row)
```

**SOLUTION:**

```
[1, 2, 3, 0, 5]
[0, 7, 0, 0, 0]
[11, 0, 13, 0, 0]
[0, 17, 0, 19, 0]
[0, 0, 23, 0, 0]
```